

KANDUKURI SAI CHARAN

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SUMMARY

Aspiring Machine Learning Engineer with strong foundations in **Python, Machine Learning, Statistics, and Data Science**. Experienced in building end-to-end machine learning solutions involving data preprocessing, exploratory data analysis, feature engineering, model selection, evaluation, hyperparameter tuning, explainability, and deployment using **Scikit-learn, Pandas, NumPy, SHAP, and Streamlit**. Hands-on experience across **classification, regression, and natural language processing (NLP)**, with a passion for developing interpretable, scalable, and data-driven AI solutions to solve real-world business problems.

EDUCATION

Bharath Institute of Higher Education and Research

Bachelor of Technology in Computer Science | CGPA: 8.5/10

Innomatics Research Labs

Data Science Program

– Coursework: Python, SQL, EDA, Power BI, Machine Learning, Deep Learning, Gen AI

Chennai, IN

2021 – 2025

Hyderabad, India

Feb 2026 – Present

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning: Scikit-learn, Classification, Regression, Model Evaluation, Cross Validation, Hyperparameter Tuning, Machine Learning Pipelines

Natural Language Processing: Text Preprocessing, Tokenization, TF-IDF Vectorization, Text Classification

Data Science: Pandas, NumPy, Matplotlib, Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization

Databases: MySQL

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Streamlit

Concepts: Object-Oriented Programming (OOP), Statistics, Data Preprocessing, Feature Selection

PROJECTS

Customer Churn Decision Support System | *Python, Scikit-learn, Pandas, NumPy, SHAP* | [GitHub](#) | [Live Demo](#)

- Built an end-to-end machine learning decision support system to predict telecom customer churn using customer demographics, service usage, contract, and billing information.
- Performed data preprocessing, exploratory data analysis, feature engineering, One-Hot Encoding, Min-Max Scaling, feature selection, model comparison, and hyperparameter tuning.
- Selected Logistic Regression based on predictive performance, interpretability, and business suitability, evaluating the model using Precision, Recall, F1-Score, and ROC-AUC to address class imbalance.
- Integrated SHAP Explainability and deployed an interactive Streamlit application providing churn prediction, customer risk assessment, executive summaries, and personalized retention recommendations.

BBC News Article Classifier | *Python, Scikit-learn, NLP, TF-IDF, Pandas, Streamlit* | [GitHub](#) | [Live Demo](#)

- Developed an end-to-end NLP-based text classification system to automatically categorize BBC news articles into predefined topics using machine learning.
- Performed text preprocessing, exploratory text analysis, tokenization, stop-word removal, text normalization, and TF-IDF vectorization to transform unstructured text into numerical features.
- Trained and compared Multinomial Naive Bayes, Logistic Regression, and Linear SVM classifiers, selecting Linear SVM based on superior Precision, Recall, F1-Score, and overall classification performance.
- Deployed an interactive Streamlit application enabling real-time news classification and prediction confidence visualization across multiple news categories.

Healthcare Cost Prediction System | *Python, Scikit-learn, XGBoost, Pandas, NumPy* | [GitHub](#) | [Live Demo](#)

- Developed an end-to-end machine learning regression system to estimate annual healthcare expenses using demographic, lifestyle, and medical attributes.
- Performed data preprocessing, exploratory data analysis, feature engineering, categorical encoding, feature scaling, and trained multiple regression models including Linear Regression, Random Forest, and XGBoost.
- Selected XGBoost Regressor based on superior predictive performance, evaluating the models using MAE, RMSE, and R^2 Score to ensure accurate healthcare cost estimation.
- Deployed an interactive Streamlit application featuring health profile analysis, risk categorization, annual cost prediction, and personalized healthcare insights for end users.